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OF BUSINESS & TECHNOLOGY KHULNA

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Project Title: AI Sales Analytics Dashboard

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Submitted to,

Name: Md. Riaz Mahmud,

Assistant Professor, Department of CSE,

Northern University of Business and Technology, Khulna, Khulna.

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1. Introduction

The AI Sales Analytics Dashboard is a full-stack, production-ready web application designed to help businesses extract actionable intelligence from their sales data through the integration of artificial intelligence, interactive visualisation, and automated reporting. Most small and medium enterprises today collect substantial volumes of transactional data but lack the analytical infrastructure to interpret it in real time, resulting in delayed decisions, undetected sales anomalies, and missed revenue opportunities (Bello, Idemudia and Iyelolu, 2024). This system directly addresses that gap by providing an intelligent, role-based platform where business users — regardless of technical background — can monitor performance, forecast trends, and interact with their data through natural language queries.

The system is built on a four-layer architecture comprising a React.js frontend, a Python FastAPI backend, a PostgreSQL relational database hosted on Supabase, and an AI service layer powered by the OpenAI or Gemini API, coordinated through LangChain for multi-turn conversational management. Each layer is independently developed and maintained by a designated team member, ensuring modularity, accountability, and clean separation of concerns throughout the development lifecycle.

1.1 Scope and Objectives

Objectives:

- To provide a secure, role-based authentication system that controls access across Admin, Manager, and Viewer user levels.
- To deliver an interactive analytics dashboard with real-time charts, KPI cards, and regional breakdowns of sales performance.
- To integrate meaningful AI features including automated insight generation, sales forecasting, anomaly detection, a natural language chatbot, and a recommendation engine.
- To support bulk CSV data ingestion alongside manual data entry for flexible data management.
- To generate AI-written monthly sales reports and support PDF and CSV export functionality.

- To deploy the complete system as a live, publicly accessible web application with full technical documentation.

Scope:

This system serves business managers, sales analysts, and administrators who need real-time visibility into sales performance without requiring specialised data science expertise. The platform is scoped to cover sales record management, multi-dimensional analytics, AI-powered intelligence features, report generation, and user administration within a single web-based interface. The system does not cover inventory procurement, accounting integration, or supply chain management, which fall outside the defined project boundary. The application will be accessible via modern web browsers on desktop, tablet, and mobile devices, and will be deployed as a live production application before the end of the semester.

2. Functional Requirements

Functional requirements define what the system must do — the specific behaviours, features, and operations the application must support (Prawira et al., 2025).

2.1 Authentication & User Management

ID	Requirement
FR-01	The system shall allow new users to register with a name, email address, and password.
FR-02	The system shall authenticate users via email and password, returning a signed JWT token upon success.
FR-03	The system shall hash all passwords using bcrypt before storing them in the database.
FR-04	The system shall enforce role-based access control, restricting features based on whether the user is an Admin, Manager, or Viewer.
FR-05	The system shall allow users to view and update their own profile information.
FR-06	The system shall allow Admin users to manage all registered user accounts.
FR-07	The system shall invalidate JWT tokens upon logout.

Table 1: Functional Requirements for Authentication & User Management

2.2 Sales Data Management

ID	Requirement
FR-08	The system shall allow authorised users to add new sales records manually through a form interface.
FR-09	The system shall allow authorised users to edit existing sales records.
FR-10	The system shall allow Admin and Manager users to delete sales records.

FR-11	The system shall support bulk data ingestion through CSV file upload.
FR-12	The system shall validate uploaded CSV files for required columns and correct data types before processing.
FR-13	The system shall allow users to filter sales records by date range, product, region, and category.
FR-14	The system shall support keyword search across all sales records.

Table 2: Functional Requirements for Sales Data Management

2.3 Dashboard & Analytics

ID	Requirement
FR-15	The system shall display KPI summary cards showing total revenue, total orders, and average order value.
FR-16	The system shall render an interactive line chart showing revenue over a selected time period.
FR-17	The system shall render a bar chart showing sales performance by product and category.
FR-18	The system shall render a pie or donut chart showing sales distribution by region.
FR-19	The system shall display a ranked table of top performing products.
FR-20	The system shall support monthly comparison views to track period-over-period growth.

Table 3: Functional Requirements for Dashboard & Analytics

2.4 AI Features

AI integration is the most critical functional layer of this system, carrying the highest evaluation weight in the course assessment. Real-world AI integration in business intelligence platforms has demonstrated measurable improvements in decision speed and analytical accuracy (Zong and Guan, 2024).

ID	Requirement
FR-21	The system shall generate an AI-written textual insight summarising current sales performance when triggered by the user.
FR-22	The system shall forecast next-month revenue using a scikit-learn regression model trained on historical sales data (Zhao, Wang and Liu, 2020).
FR-23	The system shall detect statistically unusual patterns in sales time-series data and flag them as anomalies for manager review (Choi et al., 2021).
FR-24	The system shall provide a natural language chatbot interface allowing users to query sales data conversationally via the OpenAI or Gemini API.
FR-25	The system shall generate actionable AI recommendations such as restocking alerts or promotional suggestions based on current sales trends.
FR-26	The system shall maintain conversational context across chatbot sessions using LangChain.

Table 4: Functional Requirements for AI Features

2.5 Reports & Export

ID	Requirement
FR-27	The system shall generate an AI-written monthly sales report summarising key trends, anomalies, and recommendations.

FR-28	The system shall allow users to export sales data as a CSV file.
FR-29	The system shall allow users to export generated reports as PDF files.
FR-30	The system shall send notifications when sales performance drops below a defined threshold.

Table 5: Functional Requirements for Report & Export

3. Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints the system must satisfy, independent of specific functional behaviours.

3.1 Performance

ID	Requirement
NFR-01	All standard API endpoints shall return responses within 2 seconds under normal load conditions.
NFR-02	AI-generated responses shall be returned within 10 seconds to maintain acceptable user experience.
NFR-03	Dashboard charts and KPI cards shall render within 3 seconds of page load.
NFR-04	CSV uploads of up to 10,000 rows shall be processed without system failure or timeout.

Table 6: Non-Functional Requirements

3.2 Security

ID	Requirement

NFR-05	All user passwords shall be hashed using bcrypt with a minimum salt round of 10 before database storage.
NFR-06	All API endpoints except registration and login shall require a valid JWT token for access.
NFR-07	API keys and database credentials shall be stored exclusively in server-side environment variables and never exposed to the client.
NFR-08	The system shall implement input validation on all form fields and API inputs to prevent injection attacks.
NFR-09	The system shall enforce HTTPS across all deployed endpoints.

Table 7: Non-Functional Requirements for Security

3.3 Usability

ID	Requirement
NFR-10	The interface shall be fully responsive and function correctly on desktop, tablet, and mobile screen sizes.
NFR-11	The system shall provide clear error messages for failed operations such as invalid login credentials or malformed CSV files.
NFR-12	The AI chatbot interface shall be intuitive enough for non-technical users to operate without training.
NFR-13	All charts and visualisations shall include axis labels, legends, and tooltips for clarity.

Table 8: Non-Functional Requirements for Usability

3.4 Reliability & Maintainability

ID	Requirement
NFR-14	The system shall handle API errors from OpenAI or Gemini gracefully, displaying a fallback message without crashing.
NFR-15	The backend codebase shall follow a modular folder structure separating routes, models, services, and AI logic.
NFR-16	The system shall maintain GitHub version control with meaningful commit messages and active branch management throughout development.

Table 9: Non-Functional Requirements for Reliability & Maintenance

3.5 Scalability

ID	Requirement
NFR-17	The database schema shall be designed to support future addition of new product categories, regions, and user roles without structural overhaul.
NFR-18	The backend shall be stateless and deployable across multiple instances to support horizontal scaling where required.

Table 10: Non-Functional Requirements for Scalability

4. User Roles

The system defines three distinct user roles, each with specific permissions and access boundaries.

4.1 Admin

The Admin is the highest-privileged user in the system. Admins have full access to all features including user management, sales data management, AI features, report generation, and system configuration. Only Admin users can create, edit, or deactivate other user accounts and permanently delete sales records.

4.2 Manager

The Manager role is designed for business analysts and department heads who require full visibility into sales performance without user administration privileges. Managers can add, edit, and view all sales records, access all dashboard analytics, use all AI features, generate and export reports, and receive system notifications. Managers cannot manage other users or access system-level configuration.

4.3 Viewer

The Viewer role is a read-only role designed for stakeholders who need access to reports and dashboards without the ability to modify any data. Viewers can access the main dashboard, view charts and KPI summaries, read AI-generated insights and reports, and use the AI chatbot. Viewers cannot add, edit, delete, or upload any sales data.

Table 2: Role Permission Matrix

Feature	Admin	Manager	Viewer
Register / Login	✅	✅	✅
View Dashboard	✅	✅	✅
View AI Insights	✅	✅	✅
Use AI Chatbot	✅	✅	✅
Export Reports	✅	✅	❌
Add Sales Records	✅	✅	❌
Edit Sales Records	✅	✅	❌
Upload CSV	✅	✅	❌

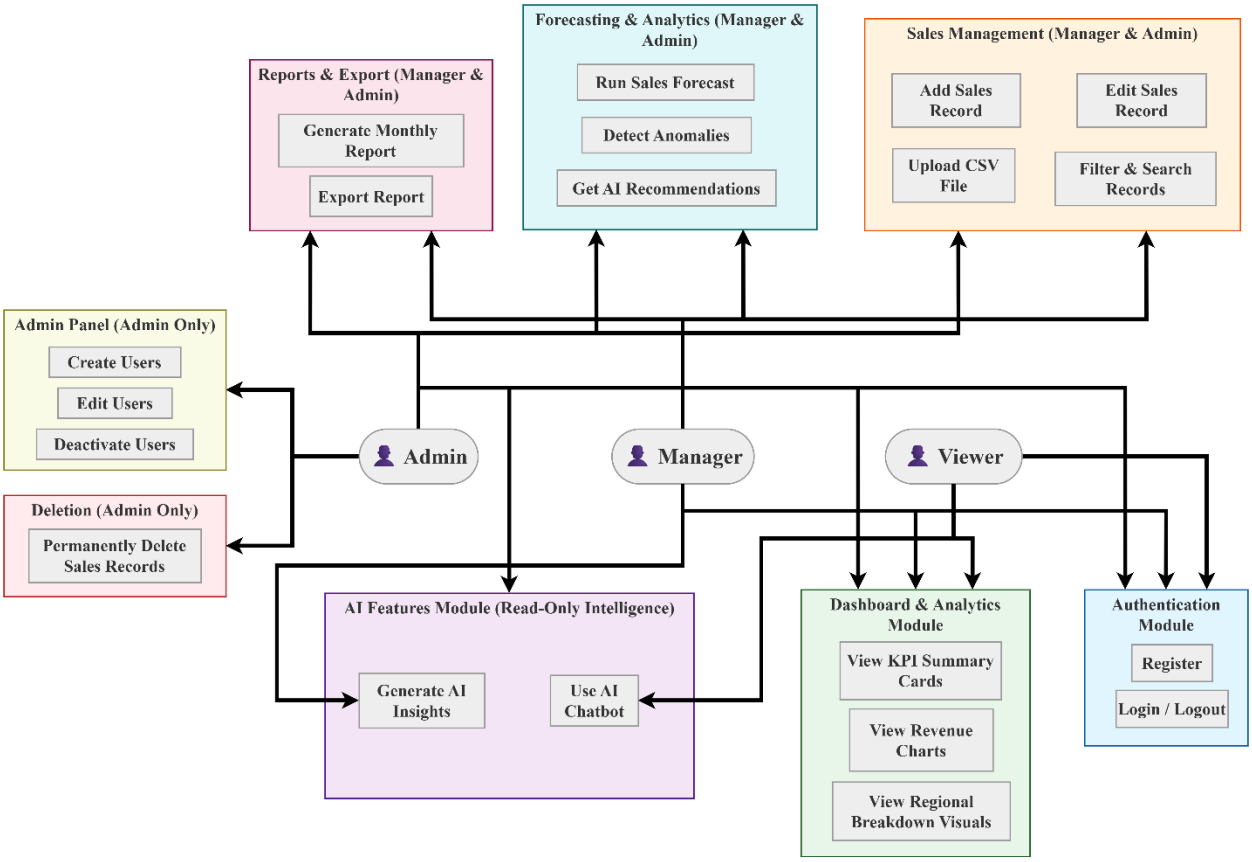
Delete Sales Records	✅	❌	❌
Manage Users	✅	❌	❌

Table 2 summarises the permission boundaries across all three roles, ensuring that data integrity is maintained and access is granted strictly on a need-to-use basis.

5. Use Cases

5.1 Use Case Diagram

The following Mermaid diagram represents the primary interactions between each user role and the system's core functional modules.



5.2 Detailed Use Cases

Use Case UC-01: User Login

Field	Detail
Use Case ID	UC-01
Title	User Login
Actor	Admin, Manager, Viewer
Precondition	User has a registered account
Main Flow	User enters email and password → System validates credentials → System returns JWT token → User is redirected to Dashboard
Alternative Flow	Invalid credentials → System returns error message → User remains on login page
Postcondition	User is authenticated and session is active

Use Case UC-02: Upload CSV Sales Data

Field	Detail
Use Case ID	UC-02
Title	Upload CSV Sales Data
Actor	Admin, Manager
Precondition	User is authenticated with Admin or Manager role
Main Flow	User selects CSV file → System validates file format and columns → System processes and stores records → System confirms successful upload
Alternative Flow	Invalid file format or missing columns → System displays validation error → No data is stored
Postcondition	Sales records are stored in the database and reflected in the dashboard

Use Case UC-03: Generate AI Sales Insight

Field	Detail
Use Case ID	UC-03
Title	Generate AI Sales Insight
Actor	Admin, Manager, Viewer
Precondition	User is authenticated and sales data exists in the system
Main Flow	User clicks Generate Insight → Backend retrieves current sales data → Data is passed to OpenAI or Gemini API with a structured prompt → AI returns written analysis → System displays insight on the AI Insights page
Alternative Flow	AI API is unavailable → System displays fallback error message → No insight is generated
Postcondition	User receives a written AI-generated summary of current sales performance

Use Case UC-04: Sales Forecasting

Field	Detail
Use Case ID	UC-04
Title	Forecast Next Month Sales
Actor	Admin, Manager
Precondition	Sufficient historical sales data exists (minimum 3 months)
Main Flow	User requests forecast → Backend retrieves historical sales records → scikit-learn regression model processes the data → Predicted revenue figure is returned → System displays forecast with confidence indicator

Alternative Flow	Insufficient data → System notifies user that more historical data is required
Postcondition	User receives a next-month revenue forecast based on historical trends

Use Case UC-05: Anomaly Detection

Field	**Detail**
Use Case ID	UC-05
Title	Detect Sales Anomalies
Actor	Admin, Manager
Precondition	Time-series sales data exists in the system
Main Flow	System automatically analyses sales time-series data → Statistical model identifies unusual spikes or drops → System flags anomalies and sends notification → Manager reviews flagged records on dashboard
Alternative Flow	No anomalies detected → Dashboard displays normal status indicator
Postcondition	Unusual sales patterns are surfaced and visible to authorised users

References

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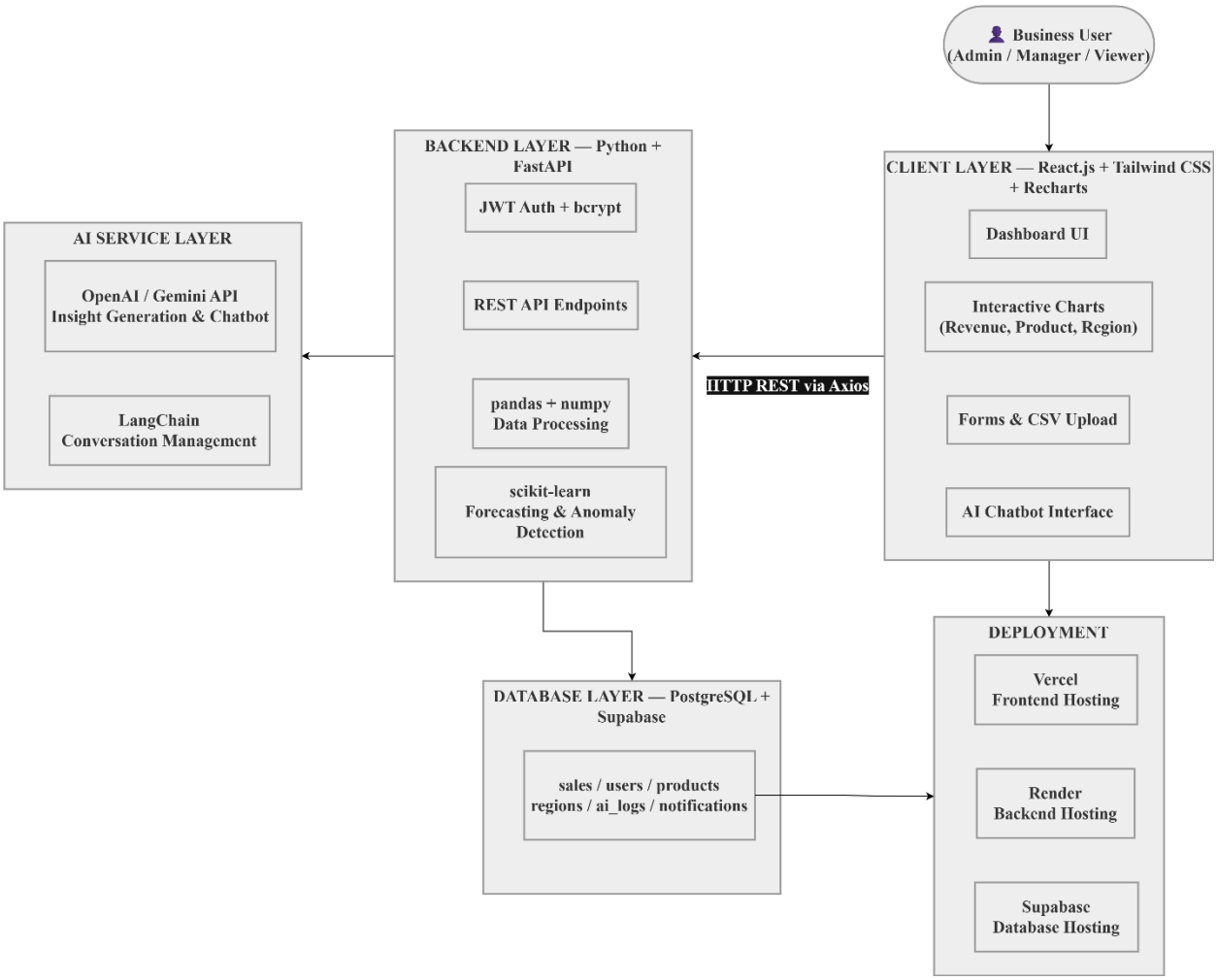
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Appendices

Appendix I: ER Diagram



Appendix II: Architecture Diagram

